SENTINEL

COMPREHENSIVE SOFTWARE TESTING REPORT

Dual-Tier Cyber-Physical Gas Safety System

Field	Details
Student	Shreedhar Sharma
Registration Number	24BPS1149
Faculty Supervisor	Dr Elakiya R
Testing tools	Selenium IDE, Jira, Apache JMeter 5.6.3
Test environment	Localhost on macOS; frontend on port 5173 and backend API on port 3001 for the recorded JMeter runs
Evidence date	21 October 2026 (as recorded in supplied execution logs and Jira export)

Prepared from the supplied test plans, execution logs, result files, Jira export, Selenium IDE suite, dashboard PDF, and screenshots.

1. Executive Summary

This report consolidates the supplied evidence for SENTINEL across three testing and test-management activities: Selenium IDE UI automation, Jira test-case tracking, and Apache JMeter API performance testing. The report distinguishes tool configuration and recorded status from independently verifiable execution evidence.

Area	Evidence observed	Conclusion
Selenium IDE	The supplied Selenium IDE .side suite contains 19 tests (UI-01–UI-13 and UI-15–UI-20). A screenshot shows the suite in the IDE with playback marked finished; no separate machine-readable Selenium run report was supplied.	Suite and UI coverage are documented; individual pass/fail totals cannot be independently reconstructed from the .side file alone.
Jira	16 of 16 exported Jira issues have Status=Done and Resolution=Done.	Jira tracking is complete at the issue-status level; some descriptions retain template text saying NOT EXECUTED, so comments and statuses must be read together.
JMeter	Four runs have supporting result/log evidence: an 80-sample read-only API sweep, PF-01, PF-02, and PF-03. All corresponding execution summaries report 0 errors.	Basic endpoint, response-time, and moderate load evidence is available. Stress, soak, input-validation, authentication, and CORS plans were supplied but no execution results for them were provided.

Overall assessment: the supplied evidence supports a successful initial functional/API and light-load test campaign. It does not establish production-scale capacity, security assurance, or safety certification. The JMeter runs are short and local, and some source artifacts contain status inconsistencies noted below.

2. Scope, Environment, and Method

System under test: SENTINEL — Dual-Tier Cyber-Physical Gas Safety System.

- Frontend: React/Vite web application, accessed locally at http://localhost:5173.
- Backend: Node.js/Express API; recorded JMeter runs target http://127.0.0.1:3001.
- Hardware context: the project status evidence supplied earlier reported a connected serial sensor device. JMeter plans used here are read-only GET requests; no claim is made that all UI scenarios were run against real hardware.
- Tools: Selenium IDE suite sentinel_merged_fixed.side; Jira CSV export for project SENTINEL Testing; Apache JMeter 5.6.3 plans, JTL result files, logs, dashboard screenshots/PDF.

Method: source artifacts were inspected for test inventory and configuration. JMeter metrics are taken from execution logs and the supplied JTL/dashboard evidence. Jira counts are based on the CSV export. Selenium coverage is enumerated from the supplied .side file and screenshots; where a distinct execution report is absent, results are labelled as not independently verified.

3. Selenium IDE UI Testing

The supplied Selenium IDE project is named “SENTINEL UI suite (full)” and contains 19 tests. The test IDs include UI-01 through UI-13 and UI-15 through UI-20; UI-14 is not present in the supplied suite. The tests cover demo scenarios, alert handling, page navigation, CSV recording/export, integration, responsiveness, latency, and accessibility.

Test case	Commands	Evidence status
UI-01 Steam is suppressed	18	Included in .side suite
UI-02 Gas leak escalates	16	Included in .side suite
UI-03 Acknowledge alarm	16	Included in .side suite
UI-04 All pages load	13	Included in .side suite
UI-05 Unknown URL redirects	4	Included in .side suite
UI-06 Demo uses simulated dispatch only	15	Included in .side suite
UI-07 Sensor fault shown	13	Included in .side suite
UI-08 ADC overflow shown	14	Included in .side suite
UI-09 Alert history logging	17	Included in .side suite
UI-10 Contributing factors drawer	16	Included in .side suite
UI-11 Caregiver view	32	Included in .side suite
UI-12 Notification settings (contacts)	11	Included in .side suite
UI-13 Demo scenario output (template)	14	Included in .side suite
UI-15 CSV recording and export	9	Included in .side suite
UI-16 Demo blocked with real device	5	Included in .side suite
UI-17 End-to-end integration	19	Included in .side suite
UI-18 Responsive layout	26	Included in .side suite
UI-19 Dashboard update latency	18	Included in .side suite
UI-20 Accessibility	18	Included in .side suite

3.1 Coverage summary

- Scenario handling: steam suppression, gas-leak escalation, sensor fault, ADC overflow, and template scenario output.
- Alert and event handling: acknowledgement, alert history, contributing-factors drawer, and end-to-end integration.
- Application features: all-pages-load, unknown-URL redirect, caregiver view, notification contacts, CSV recording/export, and real-device/demo safety guard.
- Quality checks: responsive layout, dashboard update latency, and accessibility attributes/keyboard focus.

Execution interpretation: the supplied screenshot shows the Selenium IDE suite selected with test rows visible and the playback log reporting a finished state. The .side file itself defines test steps, not a durable pass/fail report. Therefore this report does not assign a numeric Selenium pass rate without a complete runner output or exported execution report.

4. Jira Test Management and Execution Records

The supplied Jira.csv export contains 16 issues for the SENTINEL Testing project. All 16 rows have Status “Done” and Resolution “Done”. Jira Done status is treated as workflow status, not by itself as proof that every step passed.

Issue	Test case / work item	Priority	Status	Resolution
ST-4	TC-01 Execute SENTINEL End-to-End Testing	Highest	Done	Done
ST-5	TC-02: Verify Backend Connectivity	High	Done	Done
ST-6	TC-03: Verify Sensor Telemetry Updates	High	Done	Done
ST-7	TC-04: Verify Normal Operating Conditions	High	Done	Done
ST-8	TC-05: Verify Steam Scenario and False-Alarm Suppression	Medium	Done	Done
ST-9	TC-06: Verify Cooking Scenario and Nuisance Alarm Handling	Medium	Done	Done
ST-10	TC-07: Verify Gas Leak Detection and Hazard Escalation	Highest	Done	Done
ST-11	TC-08: Verify Sensor Fault Handling	High	Done	Done
ST-12	TC-09: Verify ADC Saturation Handling	High	Done	Done
ST-13	TC-10: Verify Alert Generation and Notification	Highest	Done	Done
ST-14	TC-11: Verify Backend Disconnection and Recovery	High	Done	Done
ST-17	TC-12: Verify Dashboard Recovery After Refresh	Medium	Done	Done
ST-15	TC-13: Verify Invalid API Input Handling	High	Done	Done
ST-18	TC-14: Verify API Connectivity and Telemetry Endpoints	Medium	Done	Done
ST-19	TC-15: Investigate Active Alert Consistency	Medium	Done	Done
ST-20	TC-16 — Verify Alert Acknowledgement	Medium	Done	Done

4.1 Coverage represented in Jira

- End-to-end execution and dashboard loading.
- Backend connectivity and sensor telemetry updates.
- Normal conditions, steam suppression, cooking nuisance-alarm handling, and gas-leak escalation.
- Sensor-fault and ADC-saturation handling.
- Alert generation/notification, backend disconnection/recovery, dashboard refresh recovery.
- Invalid API input, API/telemetry endpoint connectivity, active-alert consistency, and alert acknowledgement.

4.2 Data-quality note

Several Jira issue descriptions still contain template fields such as “Actual result: To be recorded after execution” or “Test execution status: NOT EXECUTED”, while their issue status is Done and later comments record TEST EXECUTION: PASS. The report therefore uses Jira comments as supporting execution notes but flags that the issue descriptions should be cleaned up before final submission so the final actual result and pass/fail status are unambiguous.

5. Apache JMeter API Performance Testing

The JMeter tests were run in non-GUI mode against the backend at 127.0.0.1:3001. The recorded plans used read-only GET endpoints, avoiding state-changing actions on the hardware-connected system.

Run	Configuration / scope	Samples	Errors	Avg (ms)	Max (ms)	Throughput (req/s)
Initial API sweep	2 users; 5 loops; 8 GET endpoints	80	0 (0.00%)	1.01	18	78.05
PF-01 Endpoint	1 user; 1 loop; 3 endpoints	3 (log)	0 (0.00%)	5 (log)	13 (log)	46.2 (log)
PF-02 Response time	10 users; 5 loops; GET /api/readings/latest	50	0 (0.00%)	2	14	5.5
PF-03 Load	50 users; 20 loops; 3 GET endpoints	3,000	0 (0.00%)	0.65 (dashboard)	16	101.8 (log); 102.02 (dashboard)

5.1 Initial API sweep (80 samples)

The supplied initial sweep JTL contains 80 successful samples across eight read-only endpoints: hardware status, latest reading, reading history, alerts, demo status, analytics configuration, caregivers, and recording status. The dashboard screenshot reports a 0.00% error rate, overall average response time of 1.01 ms, maximum of 18 ms, and throughput of 78.05 transactions/s.

5.2 PF-01 endpoint test

The uploaded PF-01 execution log reports 3 samples, 0 errors, average 5 ms, minimum 1 ms, maximum 13 ms, and throughput 46.2 requests/s. The attached pf01-results JTL contains six sample rows, so the execution log and JTL are inconsistent. Use the log result for the console run summary, but preserve this discrepancy as an evidence-quality issue.

5.3 PF-02 response-time test

The PF-02 log reports 50 samples, 0 errors, average response time 2 ms, minimum 0 ms, maximum 14 ms, and throughput 5.5 requests/s. The test uses 10 virtual users and five loops per user against GET /api/readings/latest. The test lasted approximately nine seconds.

5.4 PF-03 load test

PF-03 reports 3,000 samples, 0 errors, approximately 29 seconds elapsed, 50 users started and finished, maximum response time 16 ms, and throughput 101.8 requests/s in the execution log. The HTML dashboard reports average response time 0.65 ms, median 1 ms, 90th percentile 1 ms, 95th percentile 2 ms, 99th percentile 3 ms, and 102.02 transactions/s.

Interpretation caution: JMeter measures elapsed times at millisecond resolution; displayed 0 ms minimum or sub-millisecond averages do not mean the application literally performs work in zero time. These results apply only to the short local test runs and are not a production capacity guarantee.

5.5 JMeter plans supplied but not evidenced as executed

Plan	Purpose	Status based on supplied evidence	Caution
PF-04	Stress test (configurable 100/200/400 users)	Plan supplied; no result/log supplied	Not executed/verified in this report.
PF-05	Soak test (default 30 minutes)	Plan supplied; no result/log supplied	Not executed/verified in this report.
PF-06	Requests without credentials	Plan supplied; no result/log supplied	Requires intended authentication policy to interpret.
PF-07	Input validation POST requests	Plan supplied; no result/log supplied	Review request effects and endpoint semantics before running.
PF-08	CORS behavior	Plan supplied; no result/log supplied	Not executed/verified; browser CORS enforcement differs from a raw JMeter request.

6. Supporting Automated Quality Checks

The Jira comments for TC-16 record an automated test run with 18 tests passed and 0 failures, lint with 0 warnings and 0 errors, and a successful production build. These are reported as Jira-recorded evidence rather than newly re-executed checks for this report.

Check	Recorded outcome	Evidence note
npm test	18 passed, 0 failed	Recorded in Jira comments on TC-16.
npm run lint	0 warnings, 0 errors	Recorded in Jira comments on TC-16.
Production build	Pass	Jira comments state TypeScript compilation/build passed; the precise build output was not separately attached here.

7. Findings, Limitations, and Recommendations

- Positive: the three JMeter execution logs and the 80-sample API sweep report 0.00% errors under their recorded conditions.
- Positive: Jira export shows all 16 tracked issues in Done status with Done resolution.
- Positive: Selenium IDE suite covers 19 UI test scenarios across safety workflow, UI, export, recovery, responsiveness, and accessibility.
- Evidence gap: no standalone Selenium runner result file with a definitive pass/fail summary was supplied.
- Evidence gap: PF-04 through PF-08 have test plans but no execution result files in the supplied evidence.
- Consistency issue: Jira descriptions retain stale NOT EXECUTED/template text despite Done status and PASS comments.
- Consistency issue: PF-01 execution log reports 3 samples while the uploaded JTL has 6 rows.
- Performance limitation: runs are local and short; CPU/memory monitoring, longer-duration behavior, and higher-load resilience were not measured in the supplied results.
- Safety limitation: simulation/UI and API tests do not constitute certification of gas-leak detection accuracy, emergency dispatch behavior, or hardware safety.

8. Overall Conclusion

Based on the supplied evidence, SENTINEL has documented UI test coverage in Selenium IDE, a Jira test-management set of 16 issues marked Done, and successful initial JMeter API runs with zero reported errors. PF-03 exercised 3,000 read-only requests with 50 virtual users over approximately 29 seconds and a reported throughput of about 102 requests/s. This is useful preliminary performance evidence for a local academic project. The report does not claim that all eight JMeter plans were executed, that all Selenium cases have an independently verified pass status, or that the system is validated for production/safety-critical deployment.

9. Evidence Inventory

Artifact	Role in report
sentinel_merged_fixed.side	Selenium IDE suite definition; 19 tests.
Jira.csv	16 Jira issues and workflow statuses/comments.
pf01.jmx–pf08.jmx	JMeter test plan definitions; only PF-01–PF-03 and initial sweep have run evidence supplied.
pf01(1).log, pf02(1).log, pf03(1).log	Console execution summaries for PF-01, PF-02, PF-03.
pf01-results(1).jtl, pf02-results(1).jtl, pf03-results(1).jtl, sentinel-results(1).jtl	Raw JMeter sample data.
Apache JMeter Dashboard.pdf and screenshots	Performance dashboard and visual evidence.
SENTINEL Test Results Sheet.docx	Original planned test checklist; some expected values are targets rather than project requirements.

Appendix A — Visual Evidence

The following screenshots are included as supporting evidence. Screenshots show selected UI/Jira/JMeter states; they do not replace raw test result files.

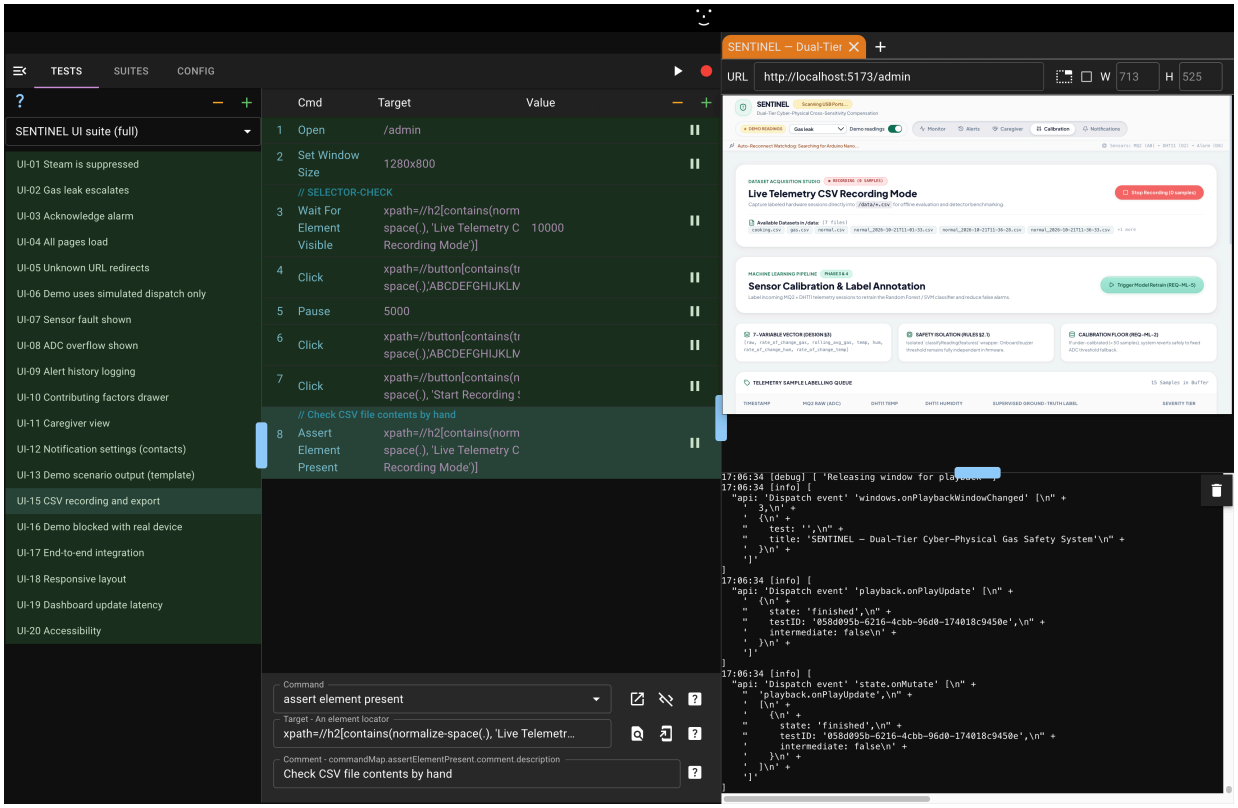


Figure A1. Selenium IDE suite and SENTINEL application view.

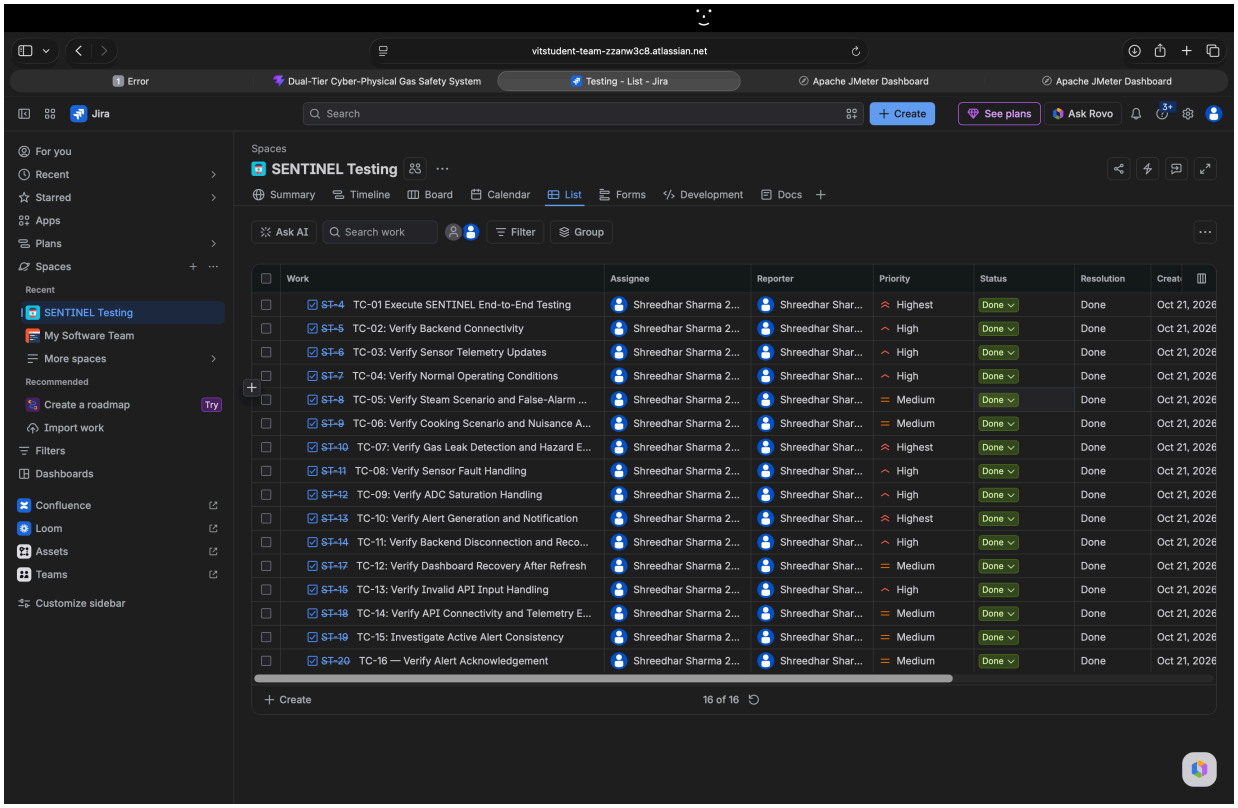


Figure A2. Jira SENTINEL Testing issue list; 16 of 16 issues shown as Done.

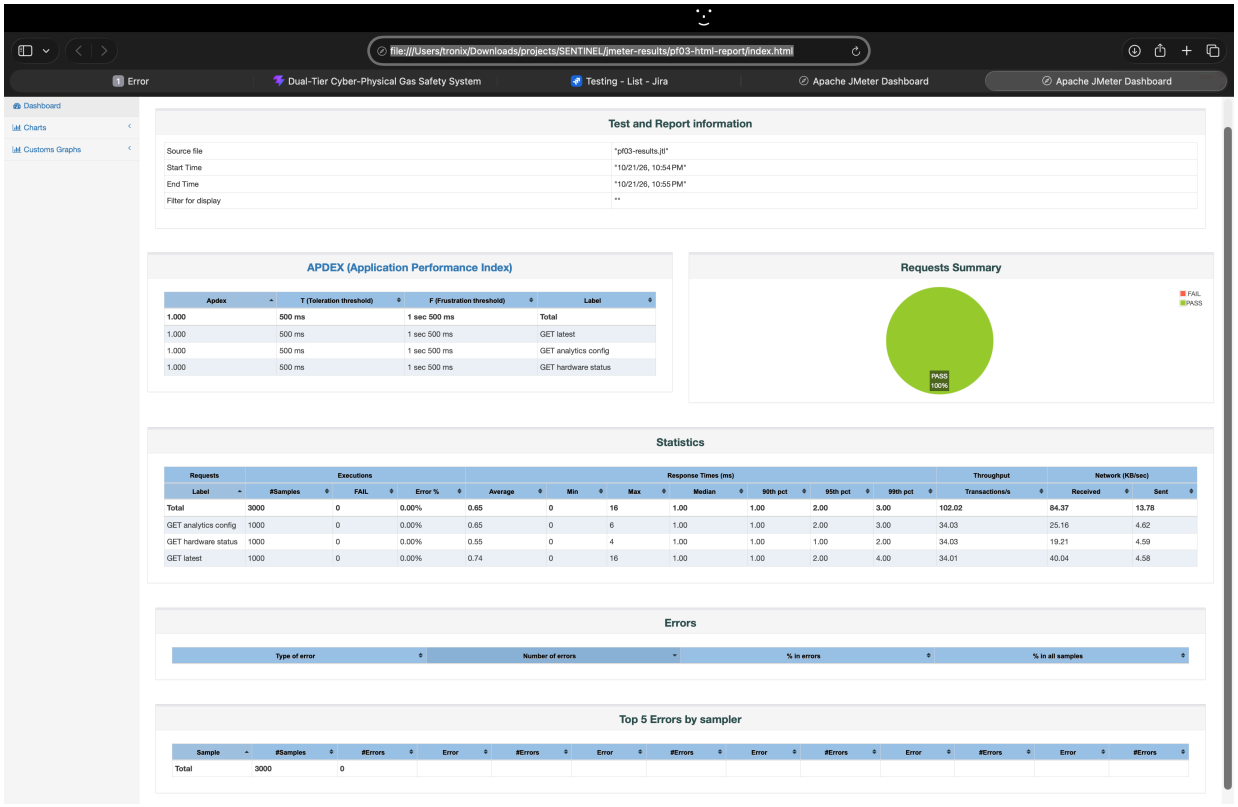


Figure A3. JMeter PF-03 dashboard statistics; 3,000 samples and 0.00% errors.

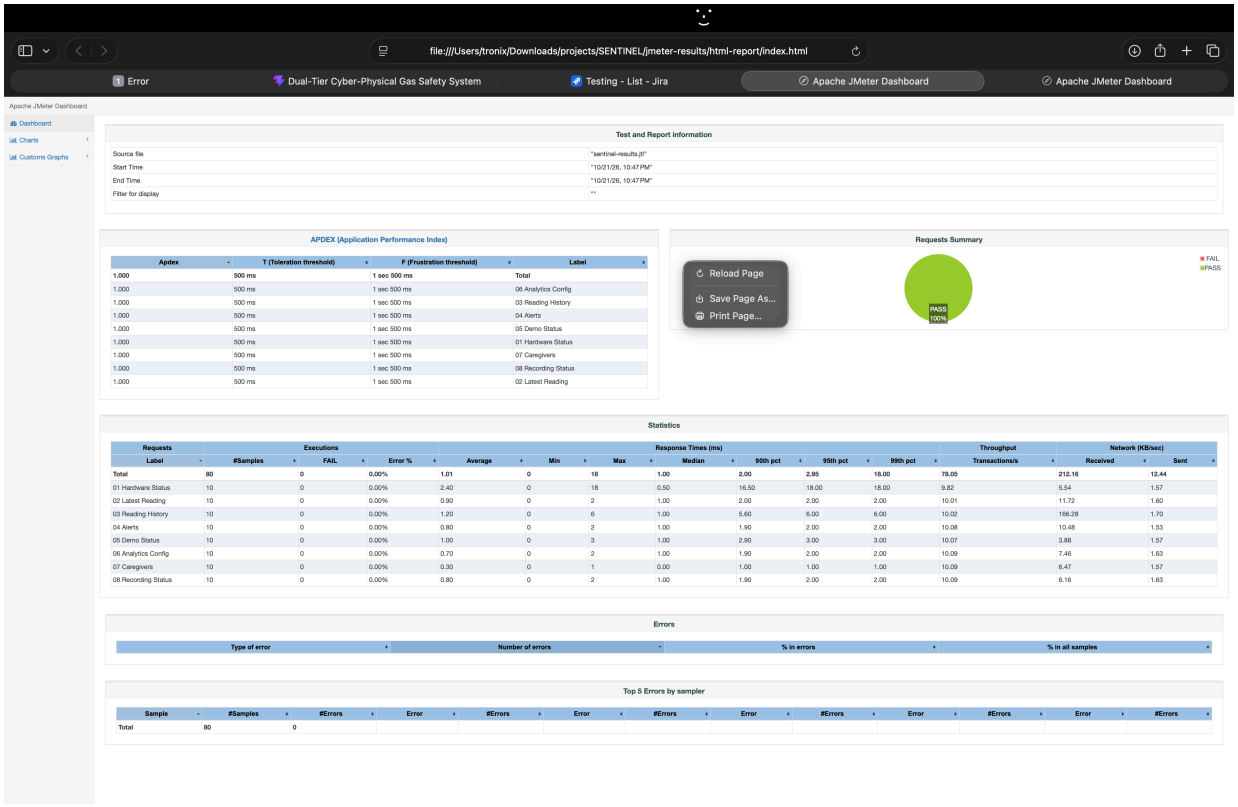


Figure A4. Initial API sweep dashboard; 80 samples across eight endpoints.

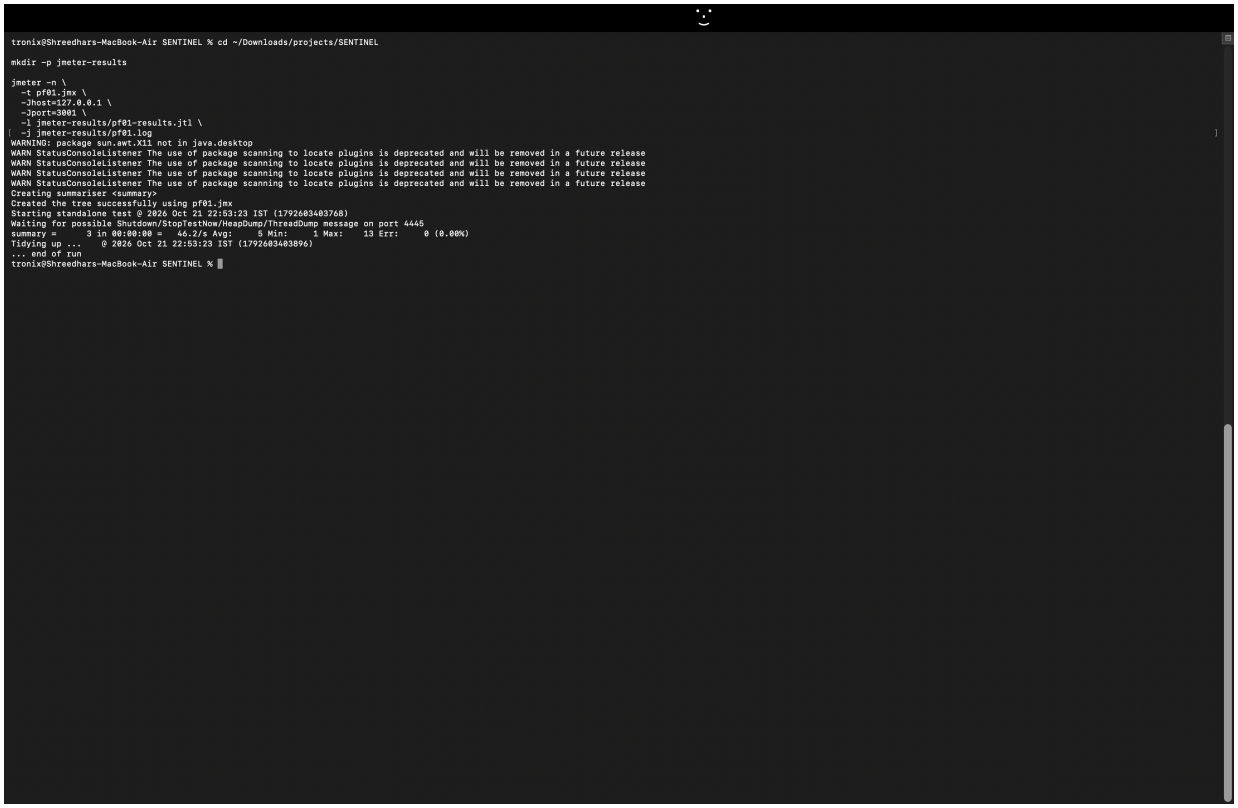


Figure A5. JMeter command-line execution evidence.

Appendix B — Selenium IDE Test Inventory

Inventory extracted from the supplied Selenium IDE .side file:

ID	Test name
UI-01	UI-01 Steam is suppressed
UI-02	UI-02 Gas leak escalates
UI-03	UI-03 Acknowledge alarm
UI-04	UI-04 All pages load
UI-05	UI-05 Unknown URL redirects
UI-06	UI-06 Demo uses simulated dispatch only
UI-07	UI-07 Sensor fault shown
UI-08	UI-08 ADC overflow shown
UI-09	UI-09 Alert history logging
UI-10	UI-10 Contributing factors drawer
UI-11	UI-11 Caregiver view
UI-12	UI-12 Notification settings (contacts)
UI-13	UI-13 Demo scenario output (template)
UI-15	UI-15 CSV recording and export
UI-16	UI-16 Demo blocked with real device
UI-17	UI-17 End-to-end integration
UI-18	UI-18 Responsive layout
UI-19	UI-19 Dashboard update latency
UI-20	UI-20 Accessibility

Appendix C — JMeter Execution Verification Screenshots

These screenshots provide visual verification of the JMeter command-line runs and generated HTML dashboards. Raw .jtl result files and execution logs remain the underlying machine-readable evidence.

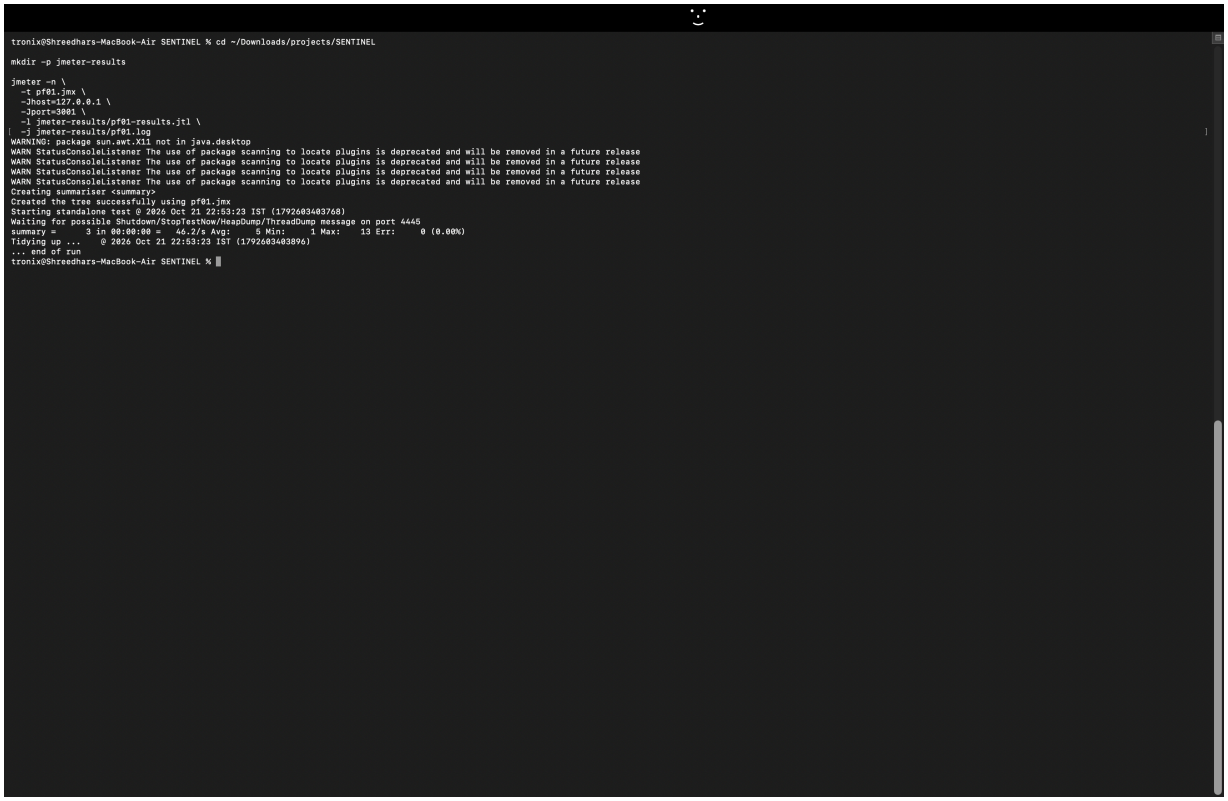


Figure C1. JMeter command-line execution output for a recorded test run.

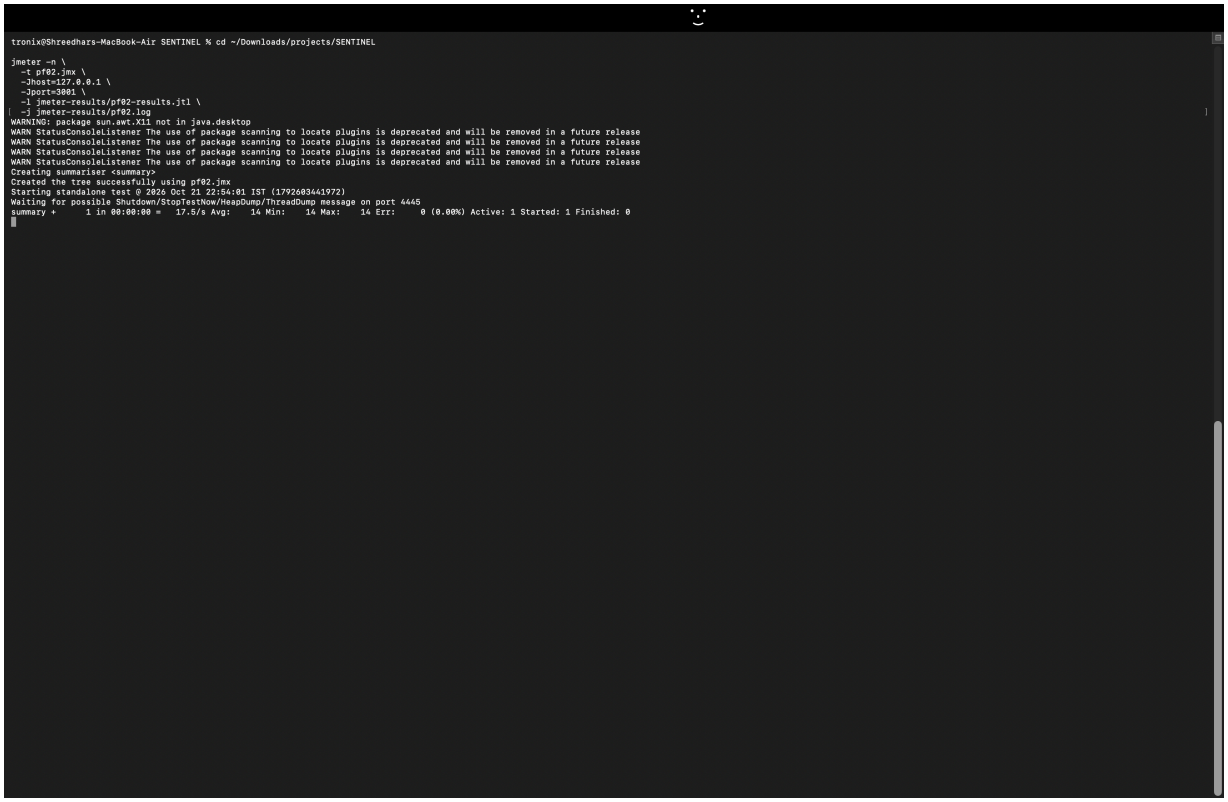


Figure C2. JMeter command-line output showing test execution summary.

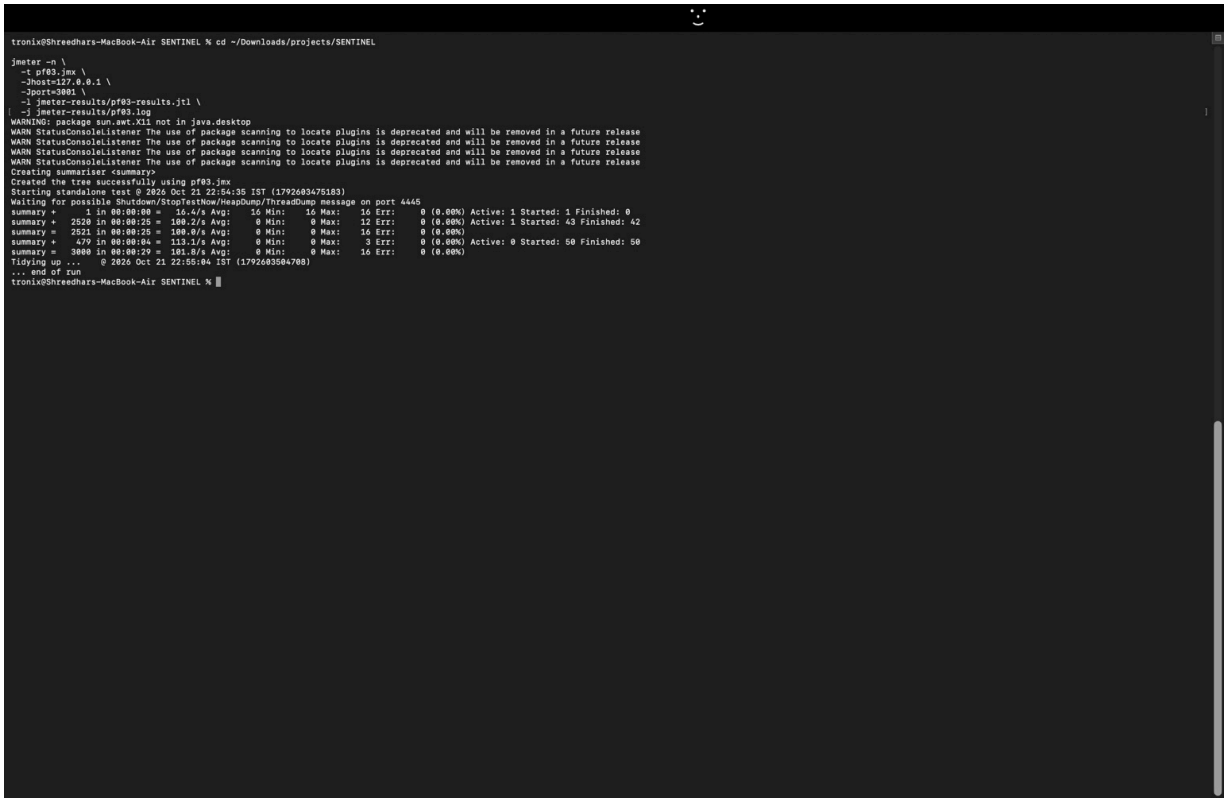


Figure C3. Terminal evidence for the PF-03 load test execution.

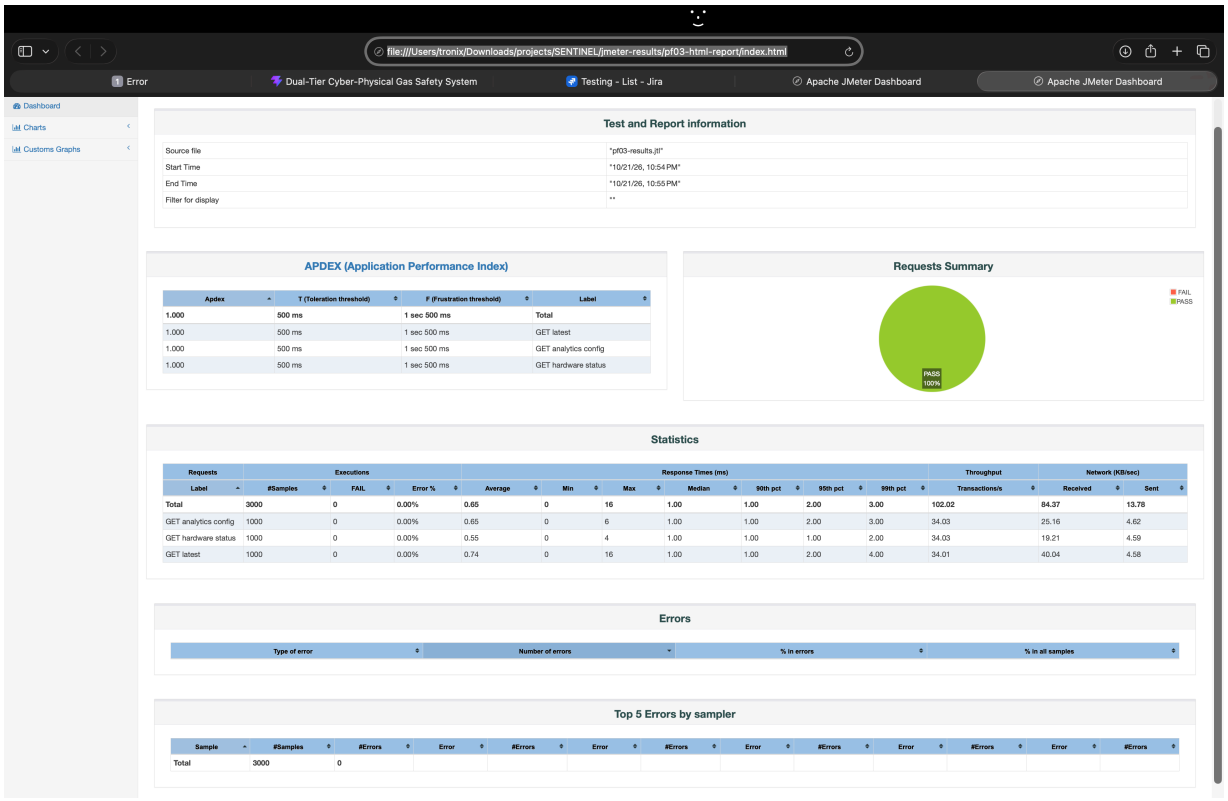


Figure C4. PF-03 HTML dashboard showing 3,000 samples, 0.00% errors, response-time percentiles, and throughput.

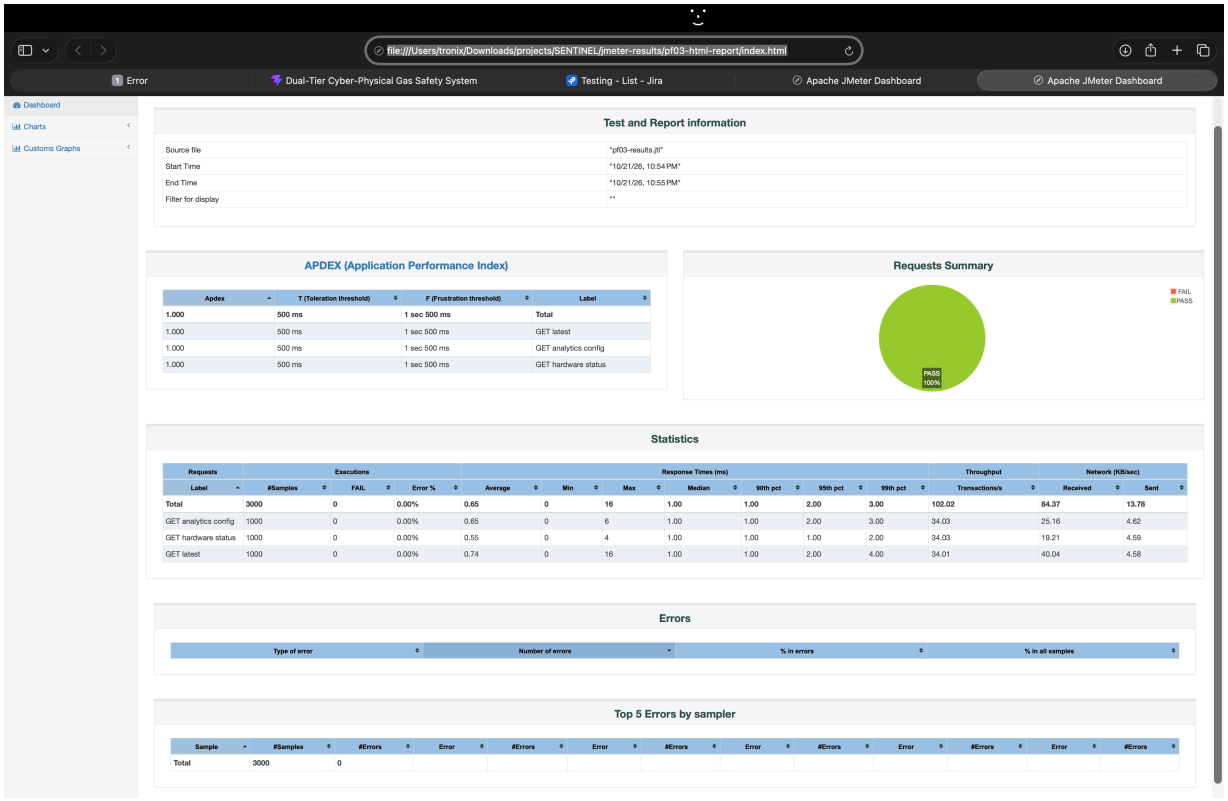


Figure C5. Additional JMeter dashboard view for the PF-03 performance run.

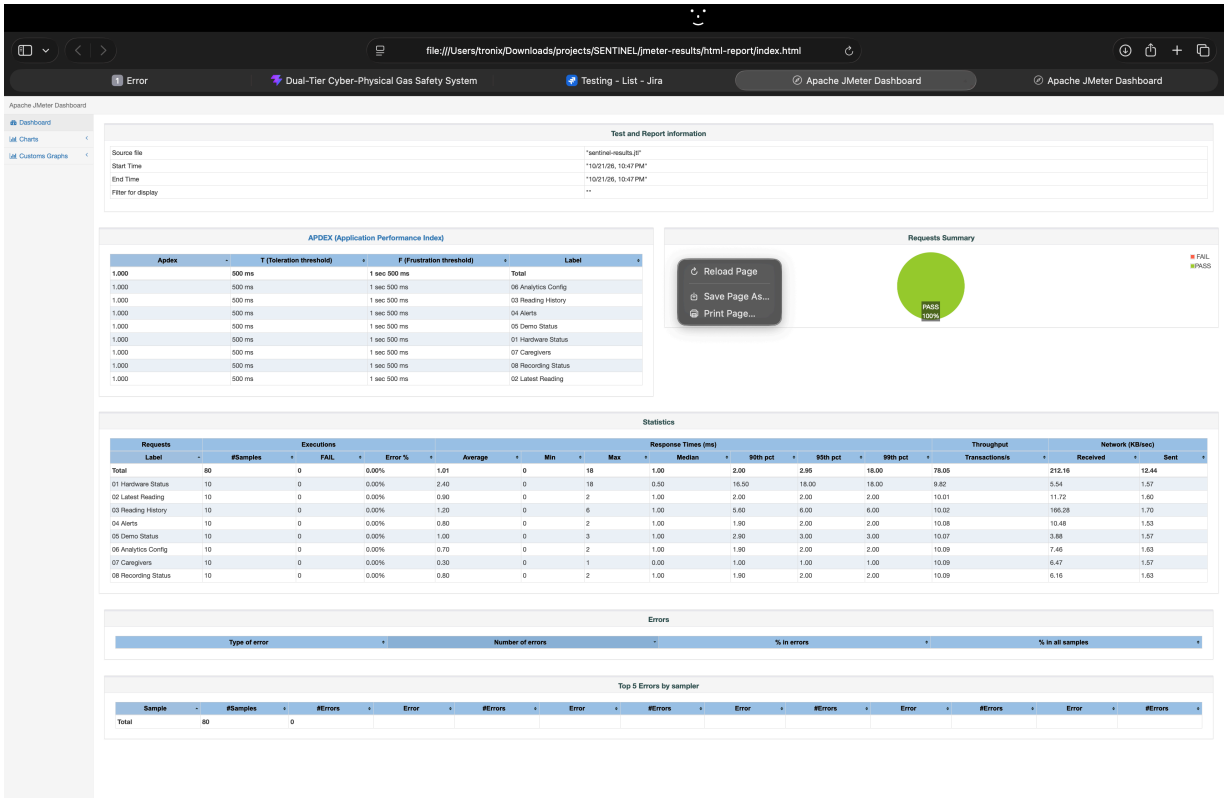


Figure C6. Initial API sweep dashboard showing 80 samples across eight read-only endpoints and 0.00% errors.